University of Potsdam
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Master Thesis
for the award of the academic degree
Master of Science (M.Sc.)
at the University of Potsdam

Forecasting the Success of Environmental and Sustainability Activities in International Development Using Language Models

Potsdam, 24 March 2026

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Abstract

Abstract in English

International aid and cooperation create a profound difference in the rate of development in growing economies, improve the lives of the world's poorest, and often safeguard the environment and materially promote sustainability. However, international aid often fails to achieve its objectives. There have been few attempts in the literature to create models to explicitly predict the success of aid activities, and none focus solely on environmental outcomes. This thesis produces a forecasting system for the overall success of international aid activities at time of evaluation from the International Aid and Transparency Initiative (IATI) database, combining classical statistical methods with modern language model techniques. Novel techniques are applied to improve prediction accuracy, including using the reasoning abilities and information gathering ability of large language models (LLMs) to improve forecasts, introducing LLM summaries of various dimensions of activity documents, using recency adjustments, and defining a novel narrative similarity grade benchmark. Overall success ratings on a scale from 1 to 6 are first assessed, then narrative forecast accuracy is measured, and finally cost-effectiveness forecasting is assessed. The full forecasting system outperforms what could be extrapolated from the stated risks in activity documents alone for overall evaluations in tests against a validation set of 300 later-starting activities in a dataset of 1,300 environmental and sustainability improving activities restricted to 4 reporting organizations. Compared to the baseline of 67 % Pairwise accuracy for the risks extrapolation, the full system reached 72 % [95% CI: 70 %, 76 %]. The forecasting system also explains 26 % [95% CI: 15 %, 37 %] of the variance in true ratings ($R^2 = 0.26$) in the validation set, compared to 10 % for a "pick the most common rating from reporting organization" baseline, and 0% for the risks extrapolation. Finally, the forecasting system predicts which activity will have a higher cost-effectiveness among random pairs of activities in the IATI database with a success probability of 60 % [95% CI: 55 %, 66 %]. This work lays the foundation to improve decision making for a wide range of initiatives and policies in developing countries and also in other data-rich institutional contexts.

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1 Introduction

Note: for this draft, the methods and results are the focus, mostly leaving out the introduction, discussion, and conclusion. Feedback is also pending before running the full analysis on the held-out test set.

This thesis analyzes the applicability and composability of machine learning techniques in forecasting the success of international aid activities using what could be known about the activities at the approval stage. The standard approach in the literature for assessing aid impact and success in environmental domains has historically involved exclusively "statistical forecasting": econometric techniques, linear regression models, and occasionally modern nonlinear models such as Random Forest (RF) or XGBoost (Goldemberg, Jordan, and Kenyon, 2025). However, a thorough assessment of methods to forecast activity success in this domain has yet to be conducted. In addition to statistical forecasting, the use of Large Language Model (LLM) implemented "judgemental forecasting" is proposed (Halawi et al., 2024 | "Approaching Human-Level Forecasting with Language Models"). Judgemental forecasting allows forecasters to use tools including Fermi estimates, intuition, and information gathering to make calibrated predictions. It is particularly appropriate when explicit reasoning can improve outcomes beyond what purely statistical correlations reveal.

The modelling system developed in this work predicts how successful interventions will be in the context of international aid activities affecting the environment, utilizing statistical forecasting, judgemental forecasting, and a mixture of the two. Metadata and PDF files from thousands of activities were obtained, and each record was separated into information about the activity available at approval and evaluation information about how successful the activity was. This data is used to answer the following research questions:

1. How do judgemental forecasting methods compare to statistical models in forecasting international aid overall success ratings and quantitative outcomes?
2. How do differing methods of combining judgemental and statistical forecasting compare in this domain?
3. What methods improve the accuracy of narrative (qualitative) forecasts in this domain?
4. What aspects of the activity available in the dataset at the beginning of the activity lead to higher or lower ratings?
5. How does forecasting aggregate cost-effectiveness compare to forecasting ratings?

To answer these questions, activity outcomes are assessed using three complementary approaches. First, the overall evaluation rating, a numerical score assigned by the funding

body at the time of evaluation, serves as the primary quantitative outcome. Second, narrative forecast similarity grades evaluate how closely an LLM-generated qualitative prediction matches the post-activity evaluation summary, providing a measure of forecast quality that is harder to game than ratings alone. Third, aggregate cost-effectiveness Z-scores, derived from quantitative results extracted from evaluation documents, offer a proxy for the relative efficiency of resource use across activities. These three outcome measures structure the empirical comparisons presented in the Results section.

This method differs in two key ways from prior literature, which has analyzed international aid evaluations. The first difference is that while many works in the literature have attempted to assess correlations between quantitative features and aid evaluation ratings, they have not focused on what knowledge would be available at approval for the activity, and do not assess out-of-time generalization of these correlations. Out-of-time generalization refers to model performance on activities that started after those used for training, as opposed to out-of-distribution generalization, which refers to performance on activities from different domains or data sources. This work assesses the out-of-time generalization ability of models. This is critical, because in order to improve aid decision making, one must assess the ability of models to forecast the outcomes of future interventions, not simply statically analyze a corpus of past activities. The second difference is that judgemental AI forecasting is implemented, as a supplement to standard statistical models.

1.1 Prior Work

Within the domain of LLM use, there have been early attempts at using them to improve decision making for the environment. A recent tool called "climsight" summarizes and aggregates information about climate adaptation and mitigation (Koldunov and Jung, 2024), but stops short of making forecasts towards adaptation. Machine learning and LLMs have been used to collect over 80,000 articles about climate adaptation and provide analysis about which areas of implementation are lacking and point out gaps in attention towards promising categories of policies (Callaghan et al., 2025).

Limited work has also been done using LLMs such as GPT-4 to serve as data sources for policy deliberation and multi-criteria assessment of climate and sustainability interventions, finding that GPT-4 is in rough agreement with the policy rankings of human experts for the expected outcomes (Bina et al., 2025). However, very little is done to improve on GPT-4's abilities, the assessment was made on only a few dozen generic policy examples, and no attempt was made to compare outcomes between these policies and real-world outcomes. Despite these limitations, the findings are promising. For multiple criteria decision making (MCDM), GPT-4 provided a useful collaborative starting point, eased the process of considering multiple criteria effectively, and aided policy deliberation on

climate change and sustainability.

One attempt which focused on specific outcomes of activities found their model using "embeddings" of LLMs were the most important features used to predict country-level sector outcomes from the World Bank World Development Indicators. They also assessed the performance of nonlinear models including the Random Forest model used in this work. However, they include features that could not be known at the beginning of the activity (e.g. actual duration). They also do not assess out-of-time generalization, instead splitting randomly within the dataset. Nor do they explicitly assess prediction performance for ratings.

2 Methods

This thesis implements an LLM-based forecasting method for ratings, a statistical model for ratings free of any LLM features, a system combining the best LLM method and the best statistical model, with the advantages of both, and finally a modified system designed to predict the cost-effectiveness of activities from quantitative results extracted from activity evaluation documents. The system is built to forecast what the evaluation results will be for thousands of IATI records containing both a pre-intervention description of the activity, and an ex-post evaluation of the results. To do so, thousands of PDFs were downloaded, ranked from most to least relevant for forecasting future outcomes or evaluating the end result of the activities, had their pages ranked and graded for relevance to the task, and had quantitative and qualitative descriptions and results transcribed into a unified format.

2.1 Data Sources

A suitable data source for this thesis required: a sufficient volume of completed environmental and sustainability activities; paired pre- and post-activity documents to enable forecasting without future information leakage (see Section 2.2); machine-readable text extractable from database records; reliable temporal metadata; and the prospect of ongoing additions to allow future replication.

Several sources were evaluated against these criteria but ultimately not used:

- **The OpenAlex publication repository of peer-reviewed evaluation documents and abstracts** (Priem, Piwowar, and Orr, 2022)
  - Although many millions of evaluations are available on OpenAlex, it proved difficult to reliably identify and de-duplicate published evaluations of environmental interventions. Furthermore, OpenAlex provides only ex-post documents

with no paired pre-activity description, making leakage prevention substantially harder.

- **The AidData database of international aid activities** (Tierney et al., 2011)
  - The AidData database has double-coded data entry. While double-coded data entry introduces less error than the LLM and regex data extraction techniques, the delay in hand-coding the ratings and results leads to relatively few recent environmental activities, and the lack of document links in AidData meant each activity record had far less extractable information. The overlap between AidData and the filtered IATI training set was less than 30% of IATI training set records. As AidData contained insufficient data for reliable forecasting on its own, and there was low overlap, merging IATI with AidData or enriching IATI with AidData was not attempted.
- **The 3ie development database of ex-post evaluations** (*3ie Development Evidence Portal | 3ie* 2025)
  - 3ie had fewer than 300 records for environmental topics, far lower than IATI, and provides only ex-post documents with no paired pre-activity description.
- **The IDEAL database of ex-post evaluations** (KfW Development Bank, 2025)
  - The IDEAL database had fewer than 300 records for environmental topics, far lower than IATI.

All activities for the thesis therefore come from the IATI database (*IATI Dashboard – IATI Activities* 2025). IATI met all selection criteria: it contains a large number of activities with substantial textual information extractable from database records, and new records are continuously added, making future replication straightforward. Crucially, IATI explicitly separates pre- and post-activity documents by design, which makes preventing temporal information leakage tractable (see Section 2.2). Records contain reliable "actual start" and "actual end" dates, and commonly also report planned start and end dates. Most records contain an overall evaluation rating on a six-point scale within the linked evaluation PDFs. Activity records also reliably mark the reporting organization. The records used for this thesis have several activity information documents uploaded near the beginning of the activity, and several years later, at least one ex-post evaluation of the activity is uploaded as well, or results of key quantitative outcomes are sometimes directly recorded in the IATI database. The status of the activity is also always reported in the database, including if it is in the planning or completion/finalization stages, which is helpful information for forecasting.

A disadvantage of the IATI database is that it is sometimes inconsistent between reporting organizations as to how the data are filled in, and the format of documents was not

always PDF, requiring conversion scripts from other formats to PDF. Many download links were also not functioning. Others required custom web-scraping scripts to extract project documents in PDF format from the original hosting websites. Planned start or end dates were often missing, leading to frequent exclusion of otherwise informative projects. Furthermore, approximately 30% of IATI activities do not have an activity category code, leading to a further exclusion of environmental or sustainability related activities that otherwise could have been included.

Country-level covariates were added from three standard sources: the World Bank governance indicators (Kaufmann and Kraay, 2024), the World Bank country policy and institutional assessment scores (World Bank, Operations Policy and Country Services, 2024), and GDP per capita (World Bank, 2024). These were included as supplementary features for the statistical forecasting models and were sometimes injected in the LLM forecasting prompts.

2.2 Temporal Information Leakage

A forecasting system designed to predict the future must not gain any information that could not be known at the start of the international aid activity. Cases where information that could not be known is used for the prediction is deemed "leakage". Preventing leakage was an important factor in the methodology. Leakage has been a significant concern in other published works where forecasting with language models has been implemented, and in multiple cases has caused invalidation of forecasting results (Paleka et al., 2025). While ex-post evaluations may provide sufficient information to describe the activity, relying on language models to completely remove information about the eventual outcome could introduce information leakage from past the start date, and was a motivation behind using IATI, which explicitly separates documents into pre- and post-activity, as opposed to OpenAlex and 3ie ex-post activity documents.

2.3 Data Filtering

Note: A longer section has been summarized here; contact the author if the original is needed.

The IATI dataset was progressively filtered from approximately 800,000 activities to 3,225 completed environment- and sustainability-related projects with both baseline and evaluation documents suitable for forecasting outcomes without future information leakage. Activities were first selected by sector codes and manual topic review, then constrained to those with downloadable documents, valid temporal ordering (baseline pdf uploaded at latest the end of first quarter of implementation period, evaluation at earliest last

quarter), and sufficient metadata reliability. Additional restrictions ensured comparable rating scales and adequate sample sizes across four reporting organizations (World Bank, BMZ/KfW/GIZ, ADB, and UK FCDO). Documents were converted to PDF and dated primarily using PDF metadata, which was validated against document content, yielding a median 22-day discrepancy. To prevent leakage from early progress reporting or misdated files, documents were screened by manual inspection, LLM-assisted checks, and automated keyword searches, resulting in the exclusion of activities showing substantive outcome progress. Further filtering removed purely legal or contractual documents, ultimately producing a curated dataset designed to reflect information realistically available near project start while preserving sufficient coverage for model training, validation, and testing.

In order to accurately extract overall success ratings for each activity and useful textual information about the project for forecasting, each PDF document was processed using the following data processing pipeline:

2.4 Data Processing

All document pages had their rotation detected, and were rotated to vertical before processing via the Gemini API. Documents with ".odt", ".doc", or ".docx" extensions were converted to PDFs with a custom script. The pages when converted to PDFs were counted and zero-page documents were excluded.

1. Ranking documents Documents were first sorted by their pdf metadata date as pre-activity and post-activity. Pre-activity documents were created at the latest $\frac{1}{4}$ of the way through the activity and post-activity documents were created at earliest $\frac{3}{4}$ of the way through, although if a date was missing, standard titles just as 'Project Information Document' or 'Ex-Post Evaluation' were used to sort the documents into pre- and post-activity. Also, the documents were given a preliminary ranking (without an LLM query). Pre-activity documents that were closest to the activity start, and the latest post-activity documents were prioritized. Documents in a 3-20 page length were also prioritized above very short or very long documents.

Documents were then given a final ranking with *gemini-2.5-flash*, ranking from most to least useful for either forecasting the outcome for pre-activity documents or evaluating the results for post-activity documents. Each activity was processed in two separate API calls, one *gemini-2.5-flash* query for pre-activity documents and a similar *gemini-2.5-flash* query for post-activity documents. In each call, the first three pages of every candidate document were uploaded alongside a single prompt containing the activity title, description, official date range, and per-document metadata (title, language, IATI document category codes, page count, and PDF metadata date). A rubric specifying the information an ideal document would contain was included to anchor the grading

scale. The model was instructed to return a single structured JSON response ranking all candidate documents from most to least relevant, assigning each a letter grade (A+ to F) and an include/exclude flag. Letter grades were selected in accordance with standard grading examples from Google Gemini documentation. Structured output was enforced via a JSON schema so that every document received exactly one rank and grade in a single response. Documents that were duplicates in a non-English language were excluded if the equivalent was available in English. For outcomes, if there were multiple progress reports, all the earlier ones were excluded and only the latest were kept in the rankings. After ranking, 2,312 documents had sufficiently informative activity information and activity evaluation documents.

2. Categorizing pages within documents The highest ranking documents then had each of their pages categorized with what information was present on them. Manual inspection determined this was more effective than semantic embedding search for long documents. To process the pages, PDFs were split into 3-page chunks. Each 3-page chunk was sent in PDF form to *gemini-2.5-flash*. Several random documents were read and categorized by-hand to understand common sections, and the resulting categories, which informed the automated categorization selection. The pages were categorized differently based on whether the document was a pre-activity or post-activity document. Categories for outcomes allowed retrieval based on whether final evaluation in quantitative or qualitative form were present on the page, deviations from plans or other types of outcomes were detailed, or if the pages were simply overviews of the activity. Specifically, the allowed categorizations were "condensed summary", "sub activities outlined", "detailed implementation plans", "broad objectives", "possible outcomes", "quantitative targets", "qualitative targets", "risks as word or numeric", "risks or dangers generally", "plans to address key risks", "positive indicators", "progress reports", "similar cases outcomes", "implementation context country", "contextual challenges", "financing details", "budget and legal", "who implements", "whether part of larger program", "partner identity or skill", "whether skin in the game", "other stakeholder engagement", or "activity monitoring details" for pre-activity document pages, and "expected outcomes", "deviation from plans", "preliminary results", "final outcomes", "delays or early completion", "over or under spending", "overview as was planned", or "unrelated to evaluation" for post-activity document pages. The two most relevant category choices among these were assigned for each page in the documents.

In order to exclude irrelevant pages, the pages were also given a second category, for post-activity document pages as "glossary", "blank page", "table of contents", "outcome evaluation", "activity description", "references", or "other", and for pre-activity document pages the same categories were options, in addition to "core activities", "theory of change", "targets", "broader context", and "preliminary results". Only one category choice among these was possible per page.

3. Extracting Ratings Two separate methods were used to extract ratings. Inspection of randomly selected pages revealed that ratings often appeared on pages with a high rating for relevance to evaluation. Therefore, the first method sends each individual post-activity document page rated above 7/10 for relevance to evaluation, or with a "quantitative targets" categorization, to *gemini-2.5-flash* to extract any overall ratings, and a second script summarized the overall ratings into a single value for the document. However, this was often insufficient to capture the overall ratings. Another "fallback" script involved a custom generated word search with approximately 500 different rephrasings of "overall rating", "final result", "synthesized score", etc, in English, and searched the PDFs directly for an exact match on those terms, prioritizing pages with one or more exact text matches of such terms. Otherwise, if such words could not be found, the earliest pages in the document which were not categorized as "blank page", "appendix", "glossary", "table of contents", "references", or "activity description" were included and *gemini-2.5-flash* was queried to extract the overall rating from the documents.

For BMZ/KfW/GIZ documents, pre-activity documents were extremely rare. For this reason, the evaluation document was treated as a pre-activity document for the purposes of forecasting activity success. Categorization for these evaluations also was via the pre-activity document method described above. When grading or summarizing the features of the evaluation document, *gemini-2.5-flash* was instructed to only describe what could have been known at the beginning of the activity, and to under no circumstances reveal the final outcome of the activity. Cursory manual inspection did not reveal any leakage, and BMZ/KfW/GIZ overall ratings were less predictable than World Bank ratings, despite this information leakage possibility.

4. Interpreting Ratings Ratings were reported both with the rating itself, and a maximum and minimum possible rating. The World Bank rating scale from 1 ("Highly Unsatisfactory") to 6 ("Highly Satisfactory") was used as the template rating, and other ratings were rescaled to match the 1 to 6 rating scale. Notably, BMZ/KfW/GIZ ratings were inverted to reach this scale. A "Satisfactory" score was considered equivalent to scores such as "successful", "On Track", or "met expectations". Scores listed as percentages or fractions were re-scaled to the 1 to 6 scale as well. In order to ensure ratings were fairly compared, only the top four most common organizations with ratings were included for training and validation of the forecasting system.

2.4.1 Outcome Cost-Effectiveness Extraction

In addition to extracting ratings, a similar approach was used with *gemini-2.5-flash* to extract quantitative cost-effectiveness-related outcomes.

Cost-effectiveness of international aid activities was defined as the ratio of measured quantitative outcomes aiming to improve the environment or sustainability to the expen-

ditures used to achieve that outcome. Quantitative outcomes were extracted by language model prompts as described below (the numerator), and the majority of international aid activities in IATI directly publish the overall activity expenditure (the denominator).

Quantitative outcomes were extracted from all pages which were categorized as containing outcomes, and marked as containing quantitative information. Unlike with ratings, extraction was not limited to the top 4 reporting organizations, as reporting ratings is more susceptible to between-reporting-organization variation and gaming than the reportable quantitative outcomes of projects (Goldemberg, Jordan, and Kenyon, 2025).

Once these PDF pages were extracted, a combination of manual examination of the extracted outcomes and a frequency analysis of bigrams was used to identify outcome variables that could be compared between projects. For each common outcome category, a list of words and phrases was compiled that would commonly match reports of these outcome variables in the description, as well as appropriate units. Each script ensures implausible or non-numeric values are dropped. Domain-specific filters also reduced false positives (e.g., wastewater context for pollution loads, water context for connections, agriculture context for yields).

Comparable Outcome Categories

Using manual inspection and frequency statistics of common bigrams, a set of outcome categories was defined that recur across evaluations and are interpretable across projects. Only those outcomes are reported that remained after ensuring they contained values for at least 10 activities after filtering to the dates of the evaluation set (2013-02-06 until 2016-06-06). The final set included:

- **Benefit/cost ratios (B/C):** benefit-cost ratio outcomes.
- **Rates of return:** economic rate of return (ERR/EIRR) and financial rate of return (FRR/FIRR), in percent.
- **Emissions reductions:** CO_2 or CO_2e reductions (total or per-year) in tonnes.
- **Water and sanitation connections:** counts of service connections, either new or repaired.
- **Energy outcomes:** installed generation capacity, in MW (or occasionally GWh where the source reported capacity in energy units).

Although pollution load removed, forest indicators, trees planted, air quality (PM2.5), clean cooking stove distribution, and agricultural yield improvement were parsed, none of these contributed sufficient counts to be useful for further outcome forecasting.

Multiple extracted values per category were aggregated by compiling all extracted outcomes in each outcome category for each activity and including them in an aggregation prompt for *gemini-2.5-flash* for aggregation into a final baseline, target, and outcome value. Where multiple extracted values conflicted, overlapped, or reflected different levels of aggregation

(e.g., component-level versus project-level figures), the model was instructed to adjudicate between them and select or combine values to reflect a single coherent project-level outcome. The model was instructed to distinguish duplicates from additive components, prioritize project-level totals over subcomponent figures, aggregate only non-overlapping quantities, and return a null response when no coherent representative value could be determined.

2.4.2 Outcome Aggregation and Cost-effectiveness Computation

Once outcomes were extracted and aggregated to project level, cost-per-unit estimates were constructed. The total disbursement for each activity reported by IATI served as the cost denominator (in USD), with the exception of benefit-cost ratios, rates of return, and agricultural yield outcomes, which are inherently unit-free or already normalised.

Splitting disbursements

Outcome-level funding splits are not published in the IATI database. To avoid implicitly treating multi-outcome activities as having multiple full budgets, a custom algorithm allocates total activity expenditure across the non-overlapping funded sub-activities and activity reports. Benefit/cost ratios and economic and financial rates of return are excluded from monetary allocation; all other outcome categories are eligible. To avoid double-counting when two indicators are alternative measurements of the same underlying result, closely related indicators are first grouped into shared conceptual buckets (e.g., protected area paired with area under management; different yield-increase measures; tree planting paired with reforested area).

Once the components are bucketed, each bucket receives an equal share of the activity's allocatable funding. Every component inside a bucket inherits that same share, so "alternative measures of the same thing" draw on one portion of the allocation rather than each taking a separate slice.

Carbon dioxide reductions are handled as a special case because they can act as a summary metric that overlaps with other mitigation outputs. If CO_2 reductions are reported without any closely linked mitigation outputs (such as improved stoves, added generation capacity, or trees planted), CO_2 reductions receive an equal share like any other bucket. If CO_2 reductions are reported alongside any of those linked outputs, they inherit the combined allocation already assigned to the CO_2-mitigating expenditures for that activity, preventing CO_2 from inflating allocated spending when it is a co-reported consequence of other outcomes.

Log transformation

Empirical inspection found that most cost-per-unit outcomes are approximately log-

normally distributed. The $\log_{10}$ transform was therefore applied to all outcome categories except benefit-cost ratios and rates of return before modelling.

2.4.3 Outcome Model Training and Evaluation

For each outcome distribution in Table 3, a random-forest regression model was trained using features and hyperparameters analogous to those used for rating forecasting, with the rating target replaced by the relevant activity-level outcome. A one-hot encoded dummy variable for the outcome category being predicted was also included. Models were trained on activities in the training split with non-missing outcomes and evaluated on the validation split. The counts in Table 3 correspond to the number of validation activities available for each outcome. Outcome categories with fewer than 10 activities in either split were excluded.

In addition, a single aggregate Z-score was computed by subtracting the training-set mean and dividing by the training-set standard deviation for each outcome. For each activity, the mean Z-score across all available outcome categories was used as the aggregate cost-effectiveness signal.

2.5 Baseline Methods

Always predict the most common rating for the reporting organization

This baseline technique provides a sanity check that more sophisticated methods are worthwhile. If this baseline outperforms more sophisticated models, it indicates that much of the methodology is overfitting on noise in the data.

Ridge regression with reporting organization and risk score

In order to assess how well calibrated overall risks are in activity documents, a ridge regression model was trained given both the LLM grade for the degree of risk assessed at the beginning of the project, as well as the dummy variable representing which organization was doing the rating. While by construction the full forecasting system also includes these features and thus with sufficient regularization must perform better than this baseline, it does provide some insight into how well we can infer the eventual rating from the risks alone in activity documents at approval. This score was only calculated for activities where the LLM was able to successfully extract the grade for how risky the project is from the pre-activity documents. This method suffers from potential inconsistency in LLMs extracting the overall risk from pre-activity documents.

***gemini-2.5-flash* summary of risks**

A separate baseline used only for assessing the forecast grades of models, is submitting

a summary of the risks identified from activity documents to the *gemini-2.5-flash-lite* grading LLM, and comparing these to proper LLM forecasts. This is a separate measure of the informativeness of risks identified in activity documents, and provides a sanity check that the LLM is not simply summarizing what the authors of activity of appraisal documents already highlight as risks. However, the lack of assertive claims about what will happen in a positive sense hinders this baseline metric relative to graded forecasts.

***deepseek-V3.2* provided only the activity summary and statistics**

Several LLM methods are introduced, closely informed by prior state-of-the-art methods in judgemental forecasting literature (Halawi et al., 2024) (Lee et al., 2025). In order to justify these methods, comparison is made to a simple baseline where all provided to the model is the activity ID, the activity title, an activity summary from *gemini-2.5-flash* of the 10 most important pages from the activity documents for forecasting the outcome, and the statistical prevalence of each outcome rating, and ask it directly to forecast the evaluation rating.

Ridge regression trained with non-LLM categories

In order to justify the addition of non-LLM categories, the baseline statistical categories used in prior literature are used and a ridge regression model is trained on the outputs. All features included improve performance on the validation set for the final forecasting system, reducing performance when removed. Sources are cited which have found meaningful predictive effects or correlations between activity success or activity outcomes and the features listed.

The non-LLM features used were:

- planned activity duration (Goldemberg, Jordan, and Kenyon, 2025)
- planned total disbursement (consultation with BMZ officers), (Bulman, Kolkma, and Kraay, 2017) (Goldemberg, Jordan, and Kenyon, 2025) (H. L. Ashton et al., 2021) (Eilers et al., 2025)
- derived features: log(planned total disbursement), planned total disbursement per year
- whether the activity is primarily loan or grant-based (Bulman, Kolkma, and Kraay, 2017)
- the one-hot encoded reporting organization (Goldemberg, Jordan, and Kenyon, 2025)
- the Country Policy and Institutional Assessment (CPIA) score from the World Bank for that country (Bulman, Kolkma, and Kraay, 2017)
- the World Governance Indicators scores (Kaufmann and Kraay, 2024) for control of corruption, government effectiveness, political stability, regulatory quality, and rule of law, as well as the mean of these values when none are median imputed (consultation with BMZ officers), (Goldemberg, Jordan, and Kenyon, 2025) (Ndikumana and

Pickbourn, 2017)

- the scope of the activity on a scale from 1-7, ranging from local to global (Vivalt, 2020)
- the log(GDP/capita) of the countries where the activity takes place weighted by the percentage of the activity performed in each country. (Goldemberg, Jordan, and Kenyon, 2025) (Bulman, Kolkma, and Kraay, 2017)
- A dummy variable for each region as defined by the World Bank (AFE, Eastern and Southern Africa; AFW, Western and Central Africa; EAP, East Asia and Pacific; ECA, Europe and Central Asia; LAC, Latin America and the Caribbean; MENA, Middle East and North Africa; SAS, South Asia) (Goldemberg, Jordan, and Kenyon, 2025) (Vivalt, 2020)
- missingness counts for important features with missingness
- interaction terms for influential variables: gdp per capita times duration

The activity start date was not used as a feature, as there was no linear pattern with regards to overall activity success over time in the training data, and adding it tended to increase overfitting and reduce out-of-time generalization. As opposed to some results in prior literature (Vivalt, 2020) (H. L. Ashton et al., 2021), whether the implementer was a government, NGO, or another category was also not found to have any improvement in model performance and was excluded.

2.6 Experimental methods

2.6.1 Statistical Forecasting Methods

Nearest Neighbor (Vector Similarity)

A similarity test was first constructed using features including countries of the activity, GDP per capita as described previously, the scope of the activity, and the implementing and funding organization ID. This similarity test was found, however, to significantly underperform compared to the semantic similarity of the *gemini-2.5-flash*-generated summary of the activity documents, so this similarity method was used instead. Similarity was first weighted proportional to its embedding semantic similarity score, and a cutoff was tested for averaging 1, 3, 7, 10, 15, and 20 nearest neighbors using the Gemini embeddings model *gemini-embedding-001*. 15 nearest neighbors was found to be the highest-performing number of neighbors using this method. The weighted average of the nearest neighbor ratings is used to predict the overall activity score.

Although the nearest neighbor method was used to collect examples for the LLM prompt, it was found that simply taking the weighted mean of 15 similar ratings performed worse than the "most common rating" method and thus was not used as an experimental method to directly forecast ratings.

Random Forest

The Random Forest (RF) method is a statistical algorithm which constructs an ensemble of decision trees which would produce the correct output on the training data, and averages those decision trees. The averaging nature of the RF algorithm reduces overfitting on the training data. The algorithm is inherently "regularized", penalizing an overly complex decision tree. The decision trees split based on value ranges of the features. By reducing the depth of the trees (the number of decision points where the decision tree splits), we can reduce the memorization of the training data from the trees, and improve generalization of the model. Each decision also only considers a random fraction of the features, allowing each tree to be more independent of each other and improving generalization further. The bootstrap method is also used to train trees, encouraging tree independence.

Extra Trees The Extra Trees (ET) method, short for Extremely Randomized Trees, is a variant of the Random Forest algorithm that constructs an ensemble of decision trees with an additional layer of randomization. While Random Forest selects the optimal split point for each randomly chosen feature subset, Extra Trees goes further by also randomizing the split thresholds, drawing cut points uniformly at random from each feature's value range and selecting the best among those random candidates. This additional randomization further decorrelates the individual trees in the ensemble, reducing overfitting at the cost of a slightly less precise fit to the data. Unlike Random Forest, Extra Trees trains each tree on the full training dataset rather than a bootstrap sample, which when ensembled with Random Forest is beneficial as it ensures the two models derive their diversity from different sources: one from varying the data each tree sees (bootstrap), and the other from randomizing split logic (ET). This makes their errors more complementary and the ensemble average more effective. As with Random Forest, predictions are made by averaging the outputs of all trees in the ensemble.

XGBoost

The XGBoost (Extreme Gradient Boosting) method is a statistical algorithm which constructs an ensemble of decision trees sequentially, where each subsequent tree attempts to correct the errors made by the previous trees. Unlike the RF which trains trees independently and averages their predictions, XGBoost builds trees iteratively, with each new tree focusing on the residual errors of the ensemble.

The algorithm incorporates both L1 and L2 regularization terms to penalize model complexity and prevent overfitting on the training data. The decision trees split based on value ranges of the features, using a splitting criterion that accounts for the gradient and hessian of the loss function.

LLM-generated scores

Seven additional features were extracted using *gemini-2.5-flash*-generated evaluation

on a score from 0 to 100 in order to enrich the activity with information from the project document PDFs from the start of the activity. These seven features were based on important aspects revealed by the literature review and consultation. One feature not introduced by others was added (the ease of targeted outcomes). Unlike technical complexity, which captures only one dimension of difficulty, or overall risk, which aggregates financial, legal, and regulatory threat sources, ease of targeted outcomes is distinct. It specifically reflects whether the activity's intended outputs are inherently measurable and achievable given typical implementation constraints—a factor theorized to independently contribute to evaluation success.

The features added for this method included the seven features and two interaction terms

- *gemini-2.5-flash*-generated evaluation on a score from 0 to 100 of:
  - how well financed the activity is (consultation with BMZ officers), (Bulman, Kolkma, and Kraay, 2017) (Goldemberg, Jordan, and Kenyon, 2025) (Eilers et al., 2025)
  - the activity integratedness within the broader activity ecosystem (consultation with BMZ officers), (Vivalt, 2020)
  - the expected implementer performance including ownership (consultation with BMZ officers), (Bulman, Kolkma, and Kraay, 2017) (Goldemberg, Jordan, and Kenyon, 2025) (H. L. Ashton et al., 2021)
  - the degree of contextual challenge (consultation with BMZ officers)
  - the overall risk level (consultation with BMZ officers), (Eilers et al., 2025)
  - the activity's overall technical complexity (Eilers et al., 2025)
  - the ease of targeted outcomes
- Additional interaction terms added
  - governance times complexity
  - expenditure times complexity

Language Model Embeddings

Statistical models are unable to explicitly capture textual information. Accordingly, this information is embedded using *gemini-embedding-001*. The input is the LLM-generated "targets" field, which summarizes the objectives of what the activity aims to accomplish. This follows (Goldemberg, Jordan, and Kenyon, 2025) in extracting the World Bank Project Development Objectives (PDO), as they are found to be very similar to the LLM-generated outputs. PDO extraction was also attempted using regex methods, but the results were found to be noisy on matching, especially on projects not from the World Bank, and did not improve prediction performance as much as embeddings on the LLM-generated targets. The LLM-extracted targets text was first normalized into a

stable canonical form (removing formatting artifacts, unescaping, splitting on separators, dropping "NO RESPONSE" tails, and deduplicating near-identical chunks). The cleaned targets text for each activity was then embedded with the *gemini-embedding-001* model, yielding a single high-dimensional vector per activity that semantically encodes activity objectives.

Following (Goldemberg, Jordan, and Kenyon, 2025), target embeddings are compressed using a two-stage dimensionality reduction pipeline. This works via Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) (Healy and McInnes, 2024). PCA projects embeddings into a few directions that preserve most of their variation by finding orthogonal directions along which the data varies most, then projecting the data onto those directions. UMAP builds a graph of nearest neighbors in high-dimensional space, then optimizes a low-dimensional embedding to preserve those relationships. PCA was first used to reduce the embedding vectors to 50 dimensions and then UMAP was fitted on the PCA outputs to produce 2D and 3D coordinate maps. The 3D embeddings (umap_x, umap_y, umap_z) were found to perform better on prediction on the validation set for forecasting activity ratings than 2D or 4D, and thus 3 dimensions were chosen. This preserves enough local topology that activities with similar targets remain near each other in the compressed space. Visual inspection also revealed that sectors with similar 2D vectors cluster around the activity environmental category, replicating the finding in (Goldemberg, Jordan, and Kenyon, 2025).

By counting the word occurrences between features higher or lower on the UMAP x,y, and z axes, it is possible to investigate what aspects are reported by the embeddings, and the common sectors which they were categorized in. Broadly, low x values correspond to forestry, agriculture, water, and management related terms while high x correspond to energy sector and financing related terms. Low y correspond to similar terms as high x, both mostly in the energy sector. High y terms related to biodiversity, wastewater, and conservation. The z axis seems to relate more to urban vs rural: Low z corresponds to wildlife and undp related terms as well as "rural", while high z corresponds to sanitation and wastewater, as well as "urban" and "city". The z axis may also correlate slightly with contextualization, where the more "cookie cutter" objectives with regards to the sector in question are typically correlated with low z values. Lastly, the z axis also has a weak negative correlations with the number of countries: low z values tend to occur in fewer countries.

This method adds 4 additional features to the model:

- umap3_x: First UMAP coordinate capturing a spectrum from forestry, agriculture, and water management objectives (low values) to energy-sector and financing-oriented objectives (high values).
- umap3_y: Second UMAP coordinate separating energy-focused activities (low

values) from biodiversity, wastewater, and conservation-oriented activities (high values).

- umap3_z: Third UMAP coordinate primarily reflecting an urban–rural axis, with lower values associated with rural and wildlife objectives and the UNDP, and higher values associated with urban sanitation and wastewater.
- a binary indicator for whether the umap embedding is missing (missing features are always imputed using the median values in the training dataset)

Budget allocation for sector clustering

A key component of the efficacy of projects is how they allocate their funding. To do so, *gemini-2.5-flash* was queried to identify the breakdown of outcome-related funding, where available in the pre-activity documents. Funding breakdowns were found to be important in prior literature (Goldemberg, Jordan, and Kenyon, 2025). However, unlike in (Goldemberg, Jordan, and Kenyon, 2025), the measure of how much the funding allocations are split between different domains (the Herfindahl–Hirschman index (HHI)) was found not to improve performance on the validation set and was not added as a feature. The allocations of funding are required to approximately sum to the total expenditure of the project, with approximately 68% of projects having their budgets identified. If no high-scoring finance or budget pages were identified in the categorization step, the first 10 pages of the highest-ranked pre-activity document were used as a fallback.

In order to make the resulting dataset of budgets usable by the statistical model, the embeddings model was used to cluster the descriptions of the subsectors into 15 sector clusters (having tried 10, 15, and 20 clusters, 15 was found to provide the optimal performance on the validation set). The allocations within each cluster are summed and reported as a fraction of the total as a feature for the statistical model.

In summary, this adds the following 16 features to the statistical model:

- 15 features, one for each sector cluster expenditure percentage (Goldemberg, Jordan, and Kenyon, 2025)
- a binary missing indicator if there is no sector cluster information in the pdf. As for other features, the result is median imputed, meaning all features are set to 0 as that is the median for all sectors in the training data.

2.6.2 LLM Forecasting Methods

NOTE: A comprehensive overview of past literature involving LLMs is included in the introduction, but has been left out of this draft.

In order to guard against leakage of information from the training, *deepseek-V3.2* was selected as the forecasting model, due to its performance on benchmarks and low cost. *gemini-3-pro* was also selected due to its performance. Lastly *gemini-2.5-flash* was selected only to assess the possibility of fine-tuning, as other more powerful models were unavailable for fine-tuning.

The LLM forecasting method was decided upon by iteratively inspecting both the quality of the response, and the overall accuracy of the forecasts made by the LLM. To generate the LLM forecasting methods, *gemini-2.5-flash* was prompted with a series of "mock forecasts", generated by *gemini-2.5-pro*. The "mock forecast" used relevant pages retrieved by ranking the categorized topics by forecast informativeness and retrieving 10 pages of the most relevant activity data and 10 pages of the most relevant evaluation data, prioritizing pages marked as "deviations from plans", "delays or early completion", or "over or under spending" with a minimum forecasting relevance score of 3/10, and otherwise returning the pages with the highest forecasting relevance score.

To generate each mock forecast, a retrieval-augmented input was constructed consisting of up to 10 baseline pages and up to 10 outcome/evaluation pages per activity. Baseline pages were selected from high-scoring passages in predefined "forecast-informative" categories (e.g., objectives, implementation plans, risks, financing details, contextual challenges, and stakeholder/implementer information), using a high relevance threshold (minimum categorization score of 9) and including nearby pages when insufficient high-scoring pages were available. Pages were selected from post-activity documents which emphasized deviations from plans (including deviations, delays/early completion, and over/under-spending), using a lower relevance threshold (minimum score of 3) and likewise included surrounding pages to reach the target count when needed. These retrieved excerpts were then merged with activity metadata (title, scope, planned start/end dates, planned financing totals when present) and brief model-generated baseline summaries (activity description and risk summary) before prompting Gemini to write a forecast from the ex-ante perspective. Importantly, the prompt required the model to end by outputting the *known* final evaluation rating for that activity (derived from the merged ratings file and converted into scale-specific text via `get_ratings_text`), while also instructing it to ground the narrative in the retrieved evaluation pages and to return "NO RESPONSE" if the evaluation excerpts did not contain sufficient justification for the assigned rating.

The most semantically relevant activities which ended approximately at or before the start of the activity being forecasted were then retrieved (see Section 2.6.1. In addition, the activity "risks" were inserted before each mock forecast, to provide context for the example. Each mock forecast was structured in a way similar to the highest performing scratchpad method from (Halawi et al., 2024). A series of features including the activity title, start date, and activity location were injected into the prompt to provide context for the activity, as well as a *gemini-2.5-flash*-generated summary.

Finally, the distribution of rating outcomes was inserted into the prompt, in order to prevent collapse towards only a few ratings.

2.6.2.1 Prompt Template and Multi-stage Prompting

The full prompt is composed of the following blocks, assembled in the order listed:

1. **Activity metadata** — title, planned start/end dates, scope, planned disbursement (USD), location(s), GDP per capita, participating and implementing organisations.
2. **Baseline description & targets** — a *gemini-2.5-flash*-generated activity description, targets summary, contextual challenge description, complexity details, integratedness description, financing summary, and implementer-performance context.
3. **Risk summary** — a *gemini-2.5-flash*-generated structured summary of key project risks.
4. **Retrieved baseline excerpts (RAG)** — additional activity-specific information retrieved by the model from the available documents and synthesised into a short passage (`{rag_synthesis_additional_info}`).
5. **KNN few-shot block** — summarised mock forecasts of the three most semantically similar training activities, one from each broad rating tier (see Section 2.6.1).
6. **Multi-stage reasoning (S1/S2)** — outputs of two prior LLM calls: reasons the activity may have been rated "Moderately Satisfactory" or worse (S1) and reasons it may have been rated "Moderately Satisfactory" or better (S2).
7. **Rating distribution prior** — a short description of the empirical distribution of rating outcomes in the training data, appended to reduce mode-collapse.

The full prompt template for the LLM Forecast is shown in Figure 2.6.2.1. LLMs in all prompts were instructed to "think like a superforecaster" (Tetlock and Gardner, 2015). Superforecasters typically outperform domain experts by maintaining broad judgement and cultivating calibrated probability estimates, rather than relying on a single strategy such as econometric analysis. This replicates prior methods which found this instruction to be the highest performing in the geopolitical domain Halawi et al., 2024.

Few-Shot Block In both methods, a k-nearest-neighbors (KNN) few-shot block of semantically similar activities in the training data is used (see Section 2.6.1 for how semantic similarity was determined). A range of nearest neighbors was evaluated. The language model was prompted to extrapolate lessons about rating scales for the most similar "Highly Unsatisfactory", "Unsatisfactory" or "Moderately Unsatisfactory", the most similar "Moderately Satisfactory", and the most similar "Satisfactory" or "Highly Satisfactory" rated examples in the training data. A selection of $k = 3$ summarized mock forecasts was found to perform better than $k = 1$, 5, or 7.

Each example activity in the few-shot block included (i) key metadata (title and, where available, location and a brief summary), (ii) a short "risks" summary, (iii) the retrospective mock forecast analysis, and (iv) the final evaluation outcome label.

Additional Prompts Two additional prompts were given, and inserted into the final forecast: (1) reasons the activity may have been evaluated as "Moderately Satisfactory" or worse, (2) reasons the activity may have been evaluated as "Moderately Satisfactory" or better.

The forecasting prompt required a structured response format that explicitly considered arguments for both lower and higher ratings, and ended with a single-line prediction. Concretely, the model was instructed to: (1) provide reasons the overall success might be rated `{midpoint_low_text}` or lower, (2) provide reasons it might be rated `{midpoint_high_text}` or higher, (3) aggregate considerations and select exactly one of the `{num_options}` outcomes, and (4) output the final forecast on the last line beginning with `FORECAST:` followed by only the chosen option.

Finally, a short description of the empirical distribution of rating outcomes in the training data was appended. This was found to reduce mode-collapse toward a narrow subset of ratings.

Box 2.1: Single-method multi-stage forecasting prompt

Stages s1 and s2 are run as separate calls, and their outputs are inserted into the final (s3) prompt via `insert_stage_s1_answer_here` and `insert_stage_s2_answer_here`. Below are the system and user prompts.

```
SYSTEM:
You are an experienced international aid decision maker with a quantitative
mindset.  Respond with a comprehensive, thorough forecast of what the overall
evaluation rating of the activity will be, from the options of {options_text}.

USER:
Forecast what the outcome will be for this activity.

### Lessons from similar activities ###
{knn_summary_text}
### End lessons ###

### Additional specific information about the activity that you summarized ###
{rag_synthesis_additional_info}
### End of additional information you summarized ###

ACTIVITY ID: {activity_id}
ACTIVITY TITLE: {activity_title}
ORIGINAL PLANNED START DATE: {planned_start}
ORIGINAL PLANNED END DATE: {planned_end}
ACTIVITY SCOPE: {activity_scope}
PLANNED TOTAL DISBURSEMENT (USD): {planned_total_disbursement_usd}
ACTIVITY LOCATION(S): {locations}
LOCATION GDP PER CAPITA, USD: {gdp_percap}
PARTICIPATING ORGANIZATIONS: {reporting_orgs}
IMPLEMENTING ORGANIZATION CATEGORY: {either "Government" or "NGO", otherwise
```

```
line not inserted}

ACTIVITY DESCRIPTION: {ChatGPT_description}
ACTIVITY TARGETS: {targets_summary}
ACTIVITY CONTEXT: {activity_context}
ACTIVITY COMPLEXITY: {complexity_details}
ACTIVITY INTEGRATEDNESS: {how_integrated_description}
FINANCING DETAILS: {finance_summary}
IMPLEMENTER PERFORMANCE CONTEXT: {implementer_performance_text}
ACTIVITY RISKS:
{risks_summary}
ACTIVITY POSSIBILITIES: {possibilities_summary}

{training-set rating distribution text}
Here are a few reasons that you said the answer might be "Moderately
Satisfactory" or worse:
{insert_stage_s1_answer_here}
Here are a few reasons that you said the answer might be "Satisfactory" or
better:
{insert_stage_s2_answer_here}

YOUR TASK:
Aggregate your considerations above.  Think like a superforecaster (e.g. Nate
Silver).  On the very last line of your response, write 'FORECAST: ' followed by
exactly one option from this rating scale with no extra words:
{options_text}

Respond only in English.
```

2.6.2.2 Inference-time Enhancements (Ensembling)

Ensembling averages many individual forecasts of the same model, prompted in slightly different ways. Ensembling was found to be a relatively weak but robust method during validation, so due to its high API cost, the ensembling method is reserved for the final held-out set, to be presented once this thesis has reached completion.

2.6.2.3 Model Adaptation (Fine-tuning / DPO)

In past work in a similar domain, fine-tuning significantly improved forecasting performance (Wen et al., 2025). An attempt was made to fine-tune *gemini-2.5-flash* using Vertex AI. To do so, Direct Preference Optimization (DPO) was used, which requires one example of a good prompt, and one example of a bad prompt. Out of a random sample of 100 activity IDs in the training data, the final forecast stage was forecasted 5 separate times using *deepseek-V3.2*. 50 pairs were used where the model forecasted one rating increment closer than another rating, as the pairs of forecasts.

It was often the case that there were multiple options for the best or worse forecast due to the limited 6-point scale. In order to choose among forecasts that were equally good or equally bad, the embeddings of the forecasts were also taken and cosine similarity was computed to the embeddings of the mock forecast, and to embeddings of the post-activity document, and these similarity scores were averaged. The most similar among equally

good ratings were chosen as the good example for the DPO training pair, and the least similar among equally bad ratings as the bad example.

Once the 50 pairs were identified, fine-tuning was conducted using default settings over 20 epochs from Vertex AI (Learning rate multiplier of 1, Adapter size of 4, Beta of 0.1).

2.6.3 Recency and LLM Adjustment Ridge Regression

The goal was to correct the RF model for temporal distribution shift (e.g., changing reporting practices, evaluation standards, portfolio composition, and macro conditions). A second goal was to incorporate any usable information from the direct LLM forecasts. Even when the input features are stable, the relationship between those features and the outcome can drift over time, so a model trained on older activities can become miscalibrated on newer ones. Furthermore, the LLM forecasts were found to be significantly correlated with prediction error in the validation set.

Residual-correction formulation

Let $\hat{y}_i^{\mathrm{RF}}$ be the RF+ET prediction for activity i, and let $\hat{y}_i^{\mathrm{LF}}$ denote the LLM Forecast. Note that the RF prediction was also averaged with the ET model, but it is still referred to as "RF" in the equations. The RF+ET residual on the i'th activity is defined as:

$$r_i := y_i - \hat{y}_i^{\mathrm{RF}}.$$

A ridge regression model was then fitted to predict residuals from a small feature vector consisting of the RF+ET prediction and (optionally) the LLM Forecast:

$$\hat{r}_i := \beta_0 + \beta_1 \hat{y}_i^{\mathrm{RF}} + \beta_2 \hat{y}_i^{\mathrm{LF}},$$

with an ℓ_2 penalty on (β_1, β_2) controlled by α, the ridge strength. The corrected prediction is:

$$\hat{y}_i^{\mathrm{corr}} := \mathrm{clip}_{[0,5]}\left(\hat{y}_i^{\mathrm{RF}} + \lambda \hat{r}_i\right),$$

where λ is a scaling factor (set to 1.0 in the experiments) and clipping enforces the valid rating range between 0 and 5.

This is a simple stacked model: the RF+ET provides the base signal, and ridge regression learns an adjustment to remove systematic residual error that appears in the recent training data.

Two separate methods were tested:

1. Recency correction (RF+ET re-calibration on recent activities). In this variant, the LLM forecast is removed entirely, fixing $\beta_2 = 0$, but an offset β_0 and scaling β_1 are

still calculated on the 150 latest training examples.

2. LLM-informed correction (recency + LLM Forecast). In this variant, the ridge model takes both the RF+ET prediction and the LLM Forecast as inputs for activities where $\hat{y}_i^{\mathrm{LF}}$ is available (the 150 latest starting activities). This allows the correction to learn a mapping from $(\hat{y}_i^{\mathrm{RF}}, \hat{y}_i^{\mathrm{LF}})$ to the residual r_i. The ridge model under this setting effectively learns how much LLM forecasts should shift the prediction from the mean, how much the RF+ET model should shift the prediction from the mean, and the recency offset correction. The only learned parameters are β_0, β_1 and β_2.

Both variants use the same 150 latest startest activities from training.

2.7 Scoring Metrics

Non-Parametric Bootstrap The non-parametric bootstrap is a method used to diversify the training data, increasing the diversity in models that are trained many times. It can be used both for ensuring methods robustly improve performance on a diversity of different training setups, and in the case of training the RF, increases independence between trees. This works by randomly sampling the same number of samples as exist in the training set, with replacement (the same training point may repeat more than once, at random). 95% confidence intervals for evaluation metrics are additionally estimated using a percentile bootstrap, repeatedly resampling prediction-outcome pairs with replacement and computing the metric on each resample, taking the 2.5th and 97.5th percentiles of the resulting distribution.

Accuracy The percent of the time the correct rating is forecasted. Non-integers are rounded to integers.

Side Accuracy The percent of correctly predicted "Satisfactory" or above vs "Moderately Satisfactory" or below (above or below 3.5). Approximately 50% of the training dataset sits above and approximately 50% sits below this boundary.

RMSE (Root Mean Square Error) Take the square of the difference between every prediction and the true value, take the mean of all such squared values, then take the square root. Measure of "average" distance. Lower is better. On a scale from 0 to 5, therefore worst possible value is 5, best possible value is zero. This method heavily penalizes predictions that are significantly incorrect.

MAE (Mean Absolute Error) Measure of "average" distance, by taking the mean of the absolute value of the residuals. Lower is better. On a scale from 0 to 5, therefore worst possible value is 5, best possible value is zero. This method does not heavily penalize predictions that are significantly incorrect.

Coefficient of Determination (R^2) R^2: Coefficient of determination. Theoretically equals zero, if we always choose the mean (however using the training set mean results in a lower score on the test set in the baseline measure below). If more than 1 regressors are included, R^2 is the square of the coefficient of multiple correlation and can be negative. Measures proportion of the variation in the dependent variable that is predictable from the independent variable. Higher is better. This method generally does not penalize outliers significantly.

Adjusted R^2 Adjusted R^2 is a version of R^2 that accounts for the number of regressors in the model. Unlike plain R^2, it penalizes adding predictors that do not meaningfully improve fit, making it more appropriate when comparing models with different numbers of features. It can decrease when irrelevant regressors are included, and it can be negative. Higher is better. While it penalizes extra parameters that may lead to overfitting, adjusted R^2 within a training set does not reflect model skill as accurately as out-of-time R^2. Adjusted R^2 is reported in this work on the training set only for comparability with prior literature, while test set $R2$ and pairwise probability are the primary results.

Pairwise Probability Pairwise probability is evaluated as the proportion of pairs of individual predictions in the validation or test set that were correctly ordered from lower to higher rating. This method is insensitive to global shifts in ratings, which may occur due to events like the COVID 19 pandemic. It also is insensitive to calibration of the spread of possible activity success ratings. Given the significant noise related to globally relevant challenges, it can represent a more achievable and informative metric than R^2 or MAE for model forecasting skill. However, because pairwise probability does not reward ties, integer ratings artificially suppress performance compared to continuous predictions.

2.7.1 Choosing Rating and Cost Effectiveness Evaluation Metrics

After consideration of several metrics, the Pairwise probability and R^2 were determined to be the most appropriate for assessing model skill in the domain of ratings and activity quantitative outcomes, while other metrics still provide useful insight. R^2 is sensitive to bias and outliers, which are important for assessing absolute predictive accuracy. However, a common use-case in aid funding decision making is comparing a pair of activities, or even a suite of many activities. In such a use-case, the Pairwise probability is more representative of what is needed. Furthermore, a weakness in R^2 is that global shifts in the ratings and outcomes due to external factors can be unpredictable, and a model may by-chance capture these shifts without any real improvement in forecasting skill. Both are clearly interpretable: always forecasting the mean value is 0, equivalent to 0% of variance explained for R^2, while 50% is equivalent to the pairwise forecast of random chance.

2.7.2 Grading Free Form Forecasts

In addition to extracting the LLM forecasted rating, *gemini-2.5-flash-lite* is also used to grade narrative textual format forecasts. In order to do so, *gemini-2.5-flash* was first used to summarize pages categorized as highly relevant to post-activity evaluations, obtaining 10 pages with a score of at least 3/10 marked as "deviation_from_plans", "delays_or_early_completion", or "over_or_under_spending", or failing that, that were graded as highly relevant to activity evaluation. If no such pages were categorized, the first 10 pages of the highest ranked activity post-activity evaluation were used. Next, for all ensembles of LLM forecasts, a grade was given for how similar the forecast was to the activity outcome using *gemini-2.5-flash-lite*. Grading was performed based on two key criteria: accurately identification of likely drivers of activity success or failure, and identification of likely outcomes being forecasted.

In accordance with the typical US grading scheme definitions, an "F" grade is defined as 55 or lower and an A+ grade as approximately 97, with other grades defined in even intervals. *gemini-2.5-flash-lite* was provided the following rubric:

Grading scale

- A+/A/A-: Excellent forecast, highly accurate, attention on key drivers, multiple major events forecasted accurately.
- B+/B/B-: Good forecast, mostly accurate. A mix of correct and incorrect, but at least one major outcome was forecasted. At least one key driver identified.
- C+/C/C-: Adequate forecast, partially accurate or reasonable. Focus was adequate. Major outcomes incorrect, but some smaller aspects were correct.
- D+/D/D-: Poor forecast, although perhaps one or two small correct things. Mostly inaccurate. Wrong focus on drivers.
- F: Failed forecast, completely wrong or unsupported.

Methodological Summary

The methodological components described above use a large quantity of unstructured data from IATI to reliably forecast international aid activity outcomes better than simplistic baseline methods. Starting from raw metadata and unstructured PDF documents, a pipeline was first developed to extract, filter, and process information available at the activity approval stage. Four forecasting approaches were evaluated: two simple approaches, and two combinations. In both cases, the combinations had superior performance to the isolated approaches.

1. A purely statistical model including Random Forest (RF) and Extra Trees (ET) were trained on structured tabular features derived from activity metadata, country-level

indicators, LLM-extracted document scores, and low dimensional language model embeddings.

2. LLM-based judgemental forecasting was implemented using multi-stage prompting, semantically retrieved few-shot training examples to generate both numerical rating forecasts and narrative qualitative predictions.

3. Third, a hybrid approach was developed in which a ridge regression is fitted to predict the residuals of the RF and ET ensemble, using the RF and ET prediction and the LLM forecast as inputs, so that systematic recency drift and LLM-derived signal can both be incorporated into a corrected final prediction.

4. Fourth, a separate hybrid approach was developed in which the more accurate statistical model overall evaluation forecast is put into the prompt for the LLM to drive narrative outcomes, which are significantly better conforming to the true narrative outcomes than the language model prediction alone.

3 Results

The sections below present results for three outcome measures: overall evaluation ratings, narrative forecast similarity grades, and aggregate cost-effectiveness Z-scores. Figure 1 provides the central empirical comparison across LLM forecasting methods and constitutes the primary evidence for the findings discussed in Section 3.1.

Three methods for measuring activity outcomes were compared. Activity ratings show only weak correlation with activity outcomes (Pearson $r = 0.07$), are frequently gamed because they influence future funding (Goldemberg, Jordan, and Kenyon, 2025) and because models with the highest rating accuracy can simultaneously produce the least accurate narratives as shown in Figure 1. The cost-effectiveness aggregate Z-score is limited by inconsistency in its component indicators across activities, difficulty of interpretation, and low sample counts. Narrative similarity grades, assessed by *gemini-2.5-flash-lite* against post-activity evaluation summaries, avoid these pitfalls: grading 300 activities costs only a few cents and takes approximately two minutes, and the underlying semantic embeddings are known to correlate with quantitative country-level outcomes (Goldemberg, Jordan, and Kenyon, 2025), making them harder to game. On this basis, narrative similarity grades are the preferred evaluation metric.

An additional aim was to assess specifically how statistical models and LLM forecasting techniques compare. Both strategies were found to have both strengths and weaknesses:

Strengths

LLM Forecasting

- Can explicitly reason and identify missing information
- General reasoning skills may transfer across domains
- Can produce text-based forecasts of specific events

Statistical Models

- Cheap and fast to iterate
- Can use large numbers of features without context limits
- Can incorporate LLM forecasts and grades as features
- Efficiently generalize over large datasets
- Easier to prevent future leakage
- Mature methodology

Weaknesses

LLM Forecasting

- Expensive and slow to iterate
- Difficult to calibrate
- Fine-tuning is expensive, and unavailable for best models
- Limited interpretability of model reasoning
- Best models are closed-source
- Risk of training-data leakage
- Limited context window constrains attention and calibration

Statistical Models

- Cannot perform explicit reasoning
- Cannot directly use rich textual or world knowledge
- More prone to overfitting dataset-specific quirks

The results from each method are described in detail in the remaining sections.

3.1 Forecasting Ratings and Narrative Forecasts with LLMs

Figure 1 summarises performance across all LLM forecasting configurations on two validation sets (Set A, $n = 300$; Set B, $n = 171$). The best narrative similarity grade is 88.5 (a "B+", corresponding to forecasts that are on average between "mostly accurate" and "highly accurate"), achieved by the RF-forced method with full context (RAG+KNN+S1+S2). The minimal-prompt baseline achieves the highest R^2 for rating prediction but the worst narrative similarity, while RF-forced methods dominate on both narrative grade and pairwise ordering probability. These patterns are consistent across both validation sets.

One research question of this thesis is how do differing methods of LLM forecasting apply in the context of forecasting international aid success ratings. More generally capable models were found to be better forecasters. The LLM's ability to predict the rating was also found not to clearly correlate with the quality of its forecast narrative. In fact, the best rating accuracy from an LLM was associated with the least accurate stated reasons. The strategy of incorporating additional information relevant to the forecasting question as well as references to similar activities does not clearly improve forecasting skill. No evidence of information leakage contaminating the ratings was found, with similar patterns of temporal correlation between statistical model average rating skill and LLM average rating skill as a function of start date. Little evidence was found that fine-tuning can improve forecasting accuracy in this domain, although a thorough assessment of the potential for fine-tuning *gemini-2.5-flash* was cost prohibitive. Language models on their own were also found to consistently underperform statistical models in forecasting skill. However, using statistical models as the drivers of LLM forecasts, statistical models were found to significantly improve the quality of LLM narrative forecasts.

In addition to extracting the LLM forecasted rating, LLM grades were also assessed. The assessed similarity between forecasts and summaries of post-activity evaluations was found to be usually correlated with improved ability to predict ratings, with the exception that very low context prompts are able to forecast accurately but for the wrong reasons (the "Baseline" method) (see Figure 1). Furthermore, it was found that when the predicted rating using the best statistical model is injected into the prompt, LLM forecasts of activities become much more similar to the summaries of their post-activity evaluations. This provides evidence that while language models can accurately forecast the reasons activities would succeed or fail if given the evaluation ratings, they generally lack the ability to correctly weigh the various factors to produce a calibrated forecast. We should be wary about using rating accuracy prediction as a sole benchmark for LLMs, given that the LLM prompt with the worst similarity to the actual rating also had the highest forecasting success. This is likely due to increased model ability when the context window is not as full - the model is better at forecasting when not overwhelmed with information

in its context window, but lacking that key information worsens its ability to pick out the best reasons for the forecasts.

While ratings are not clearly correlated with better similarity grades, RAG, KNN retrieval, and extra reasoning of the models (stages 1 and 2) consistently improve narrative textual similarity to the post-activity summary, albeit by a small margin. Simply presenting the grading model with the *gemini-2.5-flash* summary of risks of the project from the project documents lead to a forecasting score of 80.2 (between a "C+" and a "B-", corresponding to an "Adequate" forecast, with only "Adequate", rather than a good, focus on the drivers). However, the risks did not typically include any affirmative forecasts of post-activity outcomes, which may in part explain the low ratings. The best forecasting method in terms of similarity to the post-activity document summary was the forced ratings with a large amount of context (RAG+KNN) and included stage 1 and 2, achieving a grade of 88.5, which corresponds to a "B+" grade given by *gemini-2.5-flash-lite* (which sits midway between "Good" and "Excellent", meaning the forecasts were on average between "mostly accurate" and "highly accurate", with at least one major outcome forecasted, and at least one key driver of the outcome identified on average). Without any additional methods, *deepseek-V3.2* can typically achieve a "B" grade on average, while use of the statistical model or of *gemini-3-pro* can lift scores to the "B+" range. The improvement in ability to forecast overall ranking improves significantly as well.

Fine Tuning The training loss for *gemini-2.5-flash* was found to have steadily reduced over the course of the fine-tuning, reducing by 95%. This was tested on an early subset of the validation data which excluded BMZ activities. The performance results were mixed. While the R^2 performance improved modestly, it remained worse than simply guessing the mean rating in the validation set, and remained far below the performance of the more powerful *gemini-3-pro*. Furthermore, the DPO objective is specifically intended to improve pairwise performance between two pairs of forecasts. However, the Pairwise probability score was slightly worsened by fine-tuning. The cost for the fine-tuning of 20 epochs for 50 pairs was close to 140 euros. Therefore, scaling up larger training points for fine-tuning does not seem to be a promising strategy, either to reach the performance of *gemini-3-pro*, or to reach the performance of the more powerful combined statistical models.

RAG and KNN Small differences in forecasting skill were found for the addition of RAG and KNN on forecasting ratings. However, the narrative similarity scores consistently improve on the validation set when RAG and KNN context is included. This indicates that while the LLM is able to identify more important information when reasoning about the final forecast, it is insufficiently calibrated to use the additional information it has gathered to improve the forecast itself.

LLM Model Selection No consistent difference was found between *gemini-2.5-flash*

and *deepseek-V3.2* in forecasting skill. However, *gemini-3-pro* appears to be significantly better at forecasting than both, in both sets of activities achieving higher similarity grades, higher R^2, and higher Pairwise probability than other models. Two exceptions apply. Models instructed to match the statistical model's forecast outperform *gemini-3-pro* on similarity grades. Additionally, *deepseek-V3.2* using only the minimal prompt achieves a modestly better R^2 and Pairwise probability for ratings, while scoring a much worse forecast similarity score.

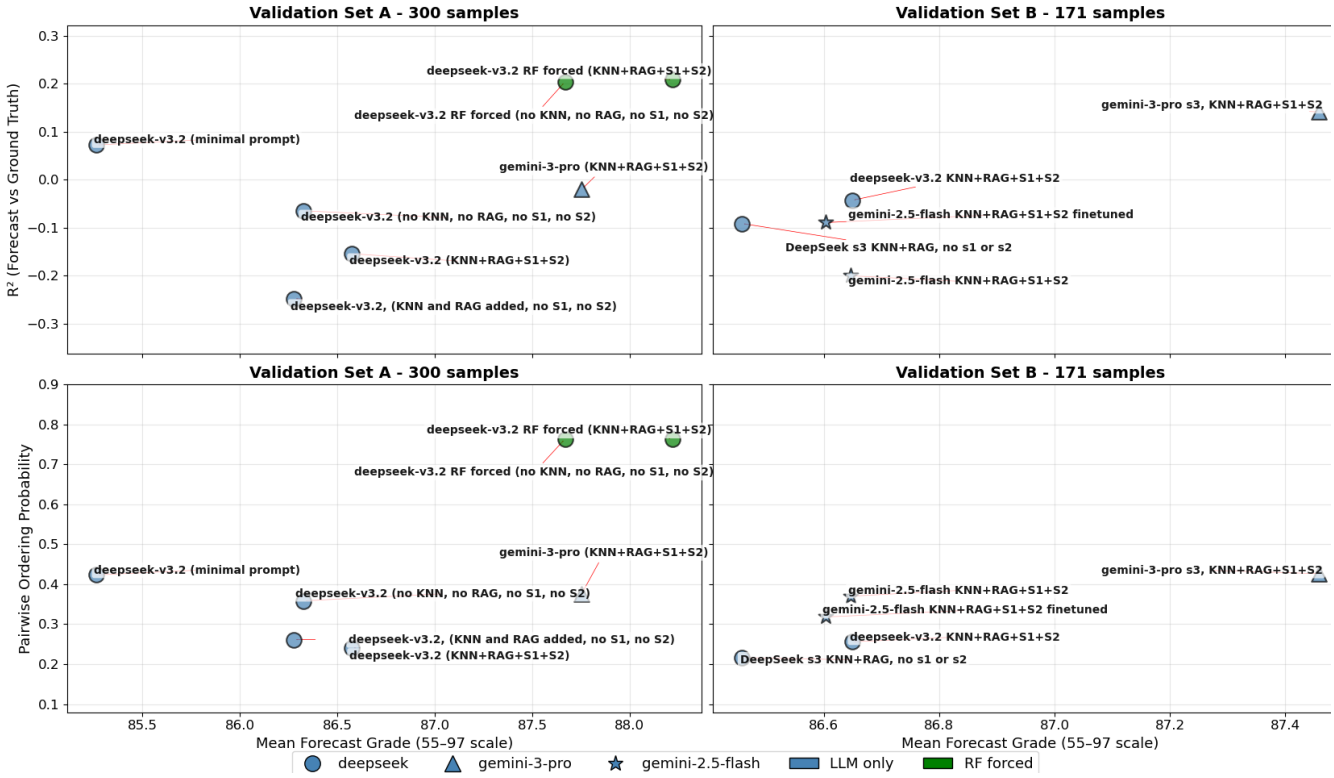

Figure 1: A comparison of LLM forecasting methods across two different validation sets (left: Set A; right: Set B). The x-axis shows the mean narrative forecast grade on a scale from 55 to 97 (higher is better; graded by *gemini-2.5-flash-lite* against the post-activity document summary). The y-axis in the top row shows R^2 between predicted and true ratings (higher is better; negative values indicate worse-than-mean predictions). The y-axis in the bottom row shows pairwise ordering probability (higher is better; 0.5 = chance). No confidence intervals are shown for individual LLM configurations. The main finding is that RF-forced methods (green) dominate on narrative grade and pairwise ordering, while the minimal-prompt baseline achieves the highest R^2 illustrating a tension between rating accuracy and narrative quality. Validation set A contains 300 activities starting after the cutoff date of February 6th, 2013 and starting before the test set cutoff of June 6th, 2016. Validation set B contains an earlier set of 171 activities that included some in the test set and some in the training set from Validation set A, and excludes BMZ activities. "KNN" refers to the retrieval of 3 similar examples into the prompt, "RAG" refers to the model additionally retrieving information it decided it was missing from the available documents and adding to the prompt, "S1" refers to additional reasoning injected into the context considering why the activity outcome may be overall unsatisfactory, "S2" refers to additional reasoning in the context considering why the activity outcome may be overall satisfactory. Because LLM forecasts are integers on a 0-5 point scale, Pairwise ordering is artificially suppressed and should be interpreted directionally.

3.2 Forecasting Ratings with Statistical Models

Overall, the forecasting system developed is capable of forecasting evaluation ratings significantly above chance on out-of-time activities. Compared to prior work (L. Ashton et al., 2023), a value consistent with their adjusted R^2 on the training set is reported. An R^2 of 0.34 and an adjusted R^2 of 0.29 are reported for within-training set correlations on primarily world bank ratings, while others report at maximum an adjusted R^2 of 0.3 (Goldemberg, Jordan, and Kenyon, 2025). No comparable out-of-time, time-ordered split analysis was identified in the literature. As expected for forecasting under data distribution shift, results on the out-of-time validation set were somewhat weaker, with an R^2 on the validation set of 0.17without LLM features for the RF+ET model, and an R^2 on the validation set of 0.26when incorporating the language model forecasting results correction + recency model. In order to compare the model with results from prior literature, adjusted R^2 on the training set of the best statistical model was calculated. It was found that the model has an adjusted R^2 of 0.24, similar to the R^2 on the validation set of 0.26, indicating overfitting is unlikely. This is also similar to the adjusted R^2 in the published literature on overall success ratings.

Several factors were found to improve model performance. The recency correction was found to provide better results than averaging LLM and statistical models, due to their widely differing forecasting skill. As the ET model performed similarly to the RF model with a slightly different algorithm, it was found that averaging the random forest model with the extra trees model also improved forecasting performance. Further successful techniques are listed below.

3.2.1 Overfitting Corrections

"Adjusted R^2" has been used in similar work to evaluate model performance in the development aid literature and penalizes overfitting by reducing the reported R^2 as a function of the number of input parameters. Mirroring similar reported methods in the literature, adjusted R^2 was calculated on the training points, in order to compare the model with results from prior literature. While this is sensitive to overfitting, it is a common practice in the development aid literature. However, adjusted R^2 within the training set was found to be highly sensitive to the specific parameters of the RF model and the subsequent degree of overfitting, such that adjusted R^2 increases to above 0.6 with default RF parameters, while performance on the validation set drops (See Table 1). It is concluded that adjusted R^2 should not be used as a measure of forecast skill.

Relative to the default `RandomForestRegressor` configuration (e.g., `n_estimators`=100, `max_depth`=`None`, `min_samples_split`=2, `min_samples_leaf`=1, `max_features`=1.0, `bootstrap`=`True`, `ccp_alpha`=0.0), the specification used here deliberately constrains

model capacity in ways that typically reduce overfitting. Trees are explicitly depth-limited (`max_depth`=14 rather than unbounded) and splits are only permitted when nodes contain substantially more data (`min_samples_split`=20 and `min_samples_leaf`=20), which smooths forecasts by limiting fine-grained partitioning of the feature space. In addition, using a smaller feature subset at each split (`max_features`=0.488) increases tree diversity and reduces variance relative to the default that considers all features. The model also uses row subsampling (`max_samples`=0.86), further reducing variance by injecting additional randomness into each tree's training set. Overall, compared to defaults, these choices trade some bias for a meaningful reduction in variance, making the fitted ensemble less susceptible to overfitting.

3.2.2 Language Model Features used for the Statistical Model

Adding the language model derived features including LLM grades (finance, integratedness, implementer_performance, targets, context, risks, and complexity), embedding features (UMAP X, Y, and Z), and sector cluster allocations to the RF raises validation R^2 from 0.17 to 0.19 ($\Delta R^2 = 0.02$; see Table 1, rows "Random Forest, no LLM features" vs. "Random Forest only").

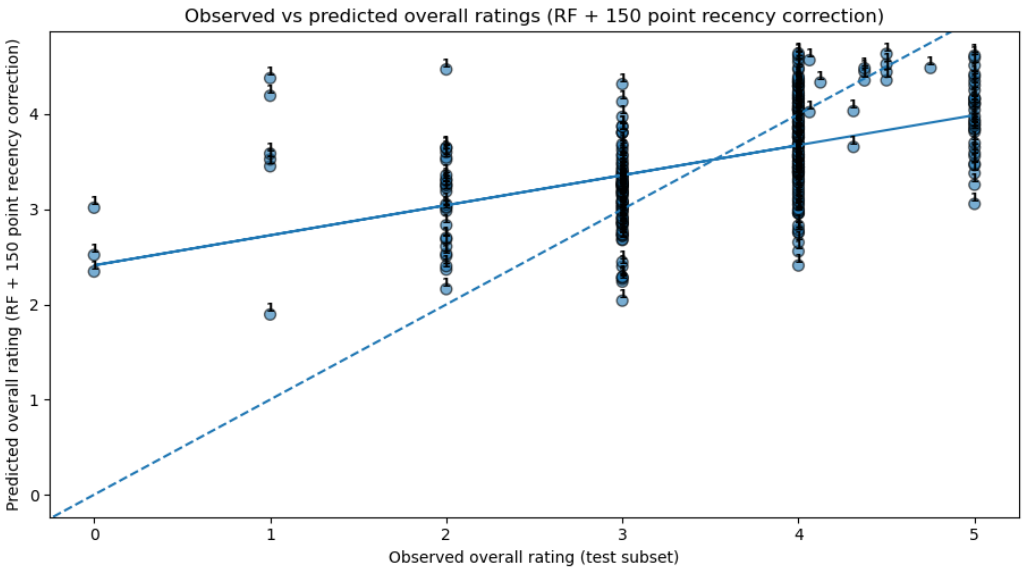

Figure 2: The scatter of observed vs predicted points on the validation set for the LLM-corrected RF prediction. $R^2 = 0.26$, Pairwise probability of 72 %.

Table 1: Validation performance in forecasting ratings across forecasting methods. Rows are sorted by ascending R^2 (higher is better). RMSE and MAE are lower-is-better, others are better if higher. Bold indicates the best value in each metric column. It was noted that MAE was better when the metric was rounded, so rounding the best model was attempted, which was found to beat the baseline metric on MAE. This is because the activity success ratings are integers, but the statistical models produce continuous prediction. The "recency correction" variants combine models using the 150 latest-starting activities in the training set for calibration/combination using a ridge regression model. XGBoost was found to underperform the RF when used as the LLM Forecast and recency combination base. The Pairwise metric is artificially suppressed for the "Mode of rep. org. score baseline" and the "RF + LLM Forecast + recency (rounded)" as these were always integers with frequent ties. The "no LLM features" RF excludes the following features: finance, integratedness, implementer_performance, targets, context, risks, complexity, umap3_x, umap3_y, umap3_z, and the 15 sector allocation features

Method	R^2 ↑	RMSE ↓	MAE ↓	Side Acc. ↑	Acc. ↑	Pairwise ↑
RF+ET + LLM Forecast + recency	**0.255**	**0.817**	0.593	**0.757**	**0.547**	0.724
RF+ET + recency	0.240	0.825	0.603	0.743	0.537	0.724
RF+ET + LLM Forecast + recency (rounded)	0.219	0.836	**0.525**	**0.757**	**0.547**	**0.758**
RF+ET (default params)	0.215	0.839	0.628	0.723	0.530	0.715
XGBoost, no LLM features	0.196	0.848	0.645	0.693	0.513	0.701
RF+ET only	0.192	0.851	0.645	0.693	0.507	0.724
Ridge GLM + Random Forest (mean)	0.185	0.854	0.649	0.690	0.523	0.707
XGBoost only	0.184	0.855	0.638	0.693	0.523	0.702
RF+ET, no LLM features	0.171	0.862	0.657	0.713	0.520	0.709
Ridge GLM	0.142	0.877	0.671	0.650	0.480	0.678
Mode of reporting-org score baseline	0.102	0.897	0.579	0.700	0.530	0.712
Ridge Baseline (risks + org only)	0.017	0.938	0.709	0.570	0.433	0.669

3.2.3 Explaining Feature Importances

SHAP (SHapley Additive exPlanations) is used to interpret model predictions by attributing changes in the output to individual features using Shapley values. For each observation, SHAP decomposes the prediction into a baseline (the mean model output) plus additive feature contributions. Each feature's SHAP value represents its average marginal contribution across all possible feature subsets.

SHAP analysis reveals that increased planned expenditure and decreased duration strongly correlate with activity success. The reporting organization dummy variable—in this case identifying World Bank activities, which have below-average ratings in this dataset—was among the most important features. The urban/rural indicator from the embeddings tended to decrease ratings, weakly indicating that less well contextualized activities performed worse. Contextual challenge weakly shifted ratings downwards, while ease of targets shifted predictions upwards. Projects spanning many countries tended to perform slightly worse.

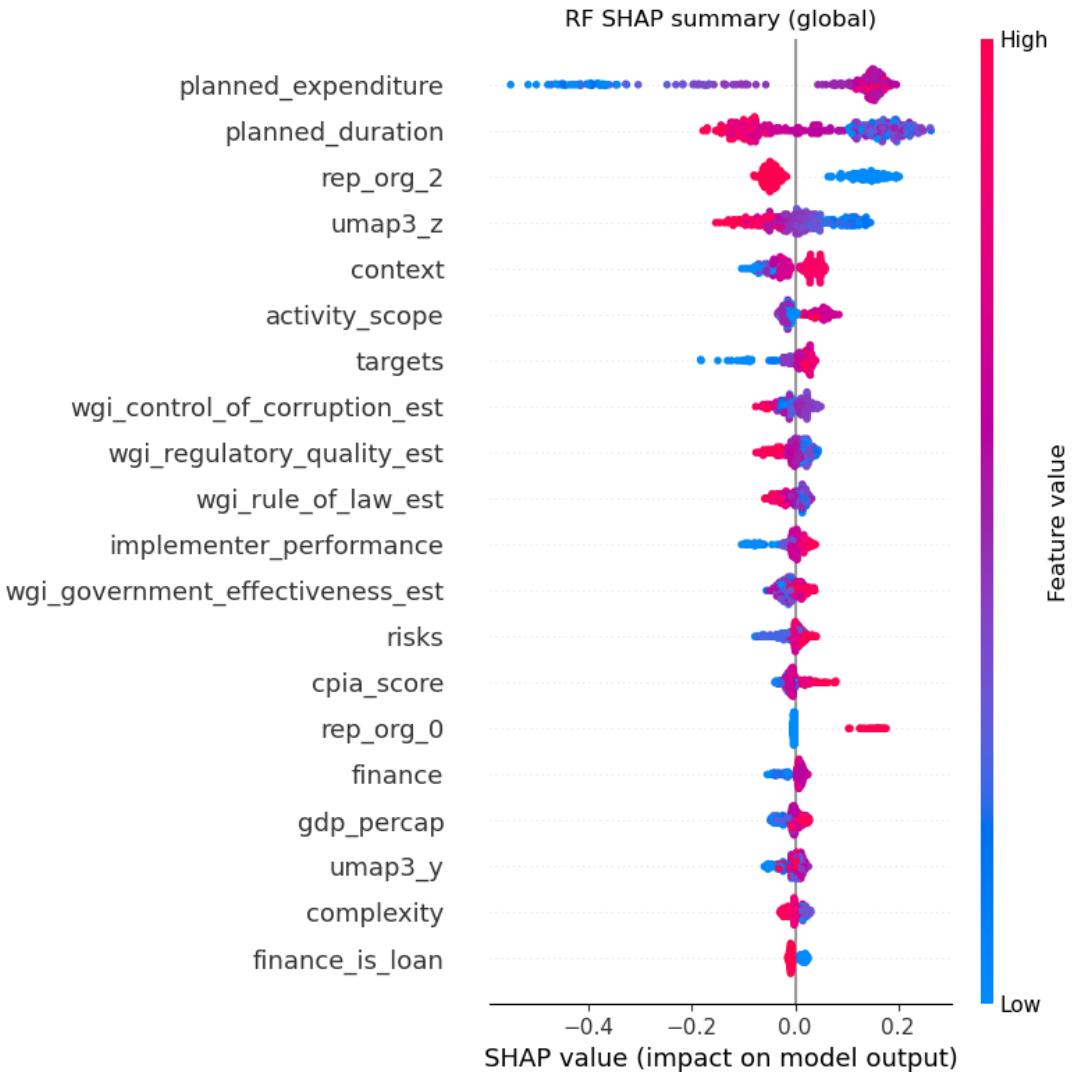

Figure 3: A SHAP analysis of ratings on the validation set from the RF. Red indicates an increase in the value of the feature, while blue indicates a below-average value. Points to the right of zero shift ratings up, points to the left of zero shift ratings down.

Table 2: Random-forest drop-one feature importance, sorted by absolute change in the prediction from the feature being increased by 1 standard deviation (Δ `pred_1sd`). Each row reports the decrease in validation R^2 and Pairwise Probability when the feature is removed (ΔR^2), along with the share of training rows that were median-imputed for that feature (`Miss` %}). The largest impacts come from planned budget and planned duration.

Feature	Miss %	$\Delta R^2 \uparrow$	ΔPairwise ↑	Δpred_1sd
planned_duration	0.0	0.0590	0.0305	-0.1028
expenditure_per_year_log	13.3	0.0051	0.0082	0.0978
planned_expenditure	8.7	0.0247	0.0210	0.0899
log_planned_expenditure	8.7	0.0326	0.0262	0.0642
activity_scope	0.0	0.0226	0.0249	0.0461
umap3_z	9.3	0.0249	0.0175	-0.0419
context	0.7	0.0316	0.0266	0.0403
cpia_score	44.7	0.0060	0.0149	0.0400
targets	0.7	0.0302	0.0243	0.0392
gdp_percap	10.0	0.0231	0.0201	0.0311
country_distance	9.3	0.0219	0.0193	0.0252
umap3_y	9.3	0.0260	0.0230	-0.0212
implementer_performance	1.0	0.0262	0.0226	0.0192
umap3_x	9.3	0.0341	0.0244	0.0183
sector_cluster_Capacity_Build	29.0	0.0442	0.0255	-0.0165
sector_distance	9.3	0.0368	0.0264	0.0154
complexity	0.3	0.0211	0.0201	-0.0106
finance	1.3	0.0155	0.0209	0.0076
governance_missing_count	0.0	0.0287	0.0227	-0.0070
risks	0.3	0.0206	0.0204	-0.0042
cpia_missing	0.0	0.0297	0.0236	-0.0033
sector_cluster_drinking_water	29.0	0.0222	0.0235	-0.0030
sector_cluster_managed_land	29.0	0.0262	0.0211	-0.0027
governance_composite	9.7	0.0061	0.0140	0.0023
region_EAP	0.0	0.0300	0.0252	0.0022
sector_cluster_Improved_transport	29.0	0.0204	0.0208	-0.0021
sector_cluster_Project_Management	29.0	0.0073	0.0193	-0.0019
sector_cluster_Institutional_cap	29.0	0.0224	0.0219	-0.0017
sector_cluster_reduced_CO2	29.0	0.0316	0.0248	0.0008
sector_cluster_Road_safety	29.0	0.0377	0.0257	-0.0007

3.3 Forecasting Cost-Effectiveness

The results from the model to predict cost-effectiveness are shown in Table 3. When dividing by each activity's disbursement for those that are marked as dollar-per-unit in Table 3, and looking at all cost-effectiveness outcomes except ratings, the correlation on Z-scores drops to 0.05, (95% CI: -0.03, 0.13) and pairwise probability of 60% (95% CI: 54%, 67%) ($N_{\text{val}} = 299$), compared to an R^2 of 0.10 (95% CI -0.0982, 0.2625) and a pairwise probability of 68% (95% CI: 0.62, 0.74) (n_val=84) when forecasting the Z-score for CO_2 emission reductions, generation capacity, and water connections. While pairwise ordering ability for cost-effectiveness outcomes is statistically significant, the overall ability is weak and appears to be largely a function of expenditure and sector clusters.

Interpreting these results, although it is likely that cost-effectiveness bests a simple baseline of predicting the mean of the aggregate cost-effectiveness Z-score for the validation set with 95% confidence (see Figure 3), this absolute prediction accuracy cannot be distinguished from the baseline with 95% confidence. Despite this weak performance in absolute accuracy, ranking cost-effectiveness performs modestly better. The model is able to distinguish cost-effectiveness between pairs of activities, at a rate of 60% accuracy. The model also beats the ranking performance of a random ordering (50%) with 95% confidence.

The top two features of the model are UMAP x axis (higher x-axis values reduce cost-effectiveness), and the planned duration (longer durations increase cost-effectiveness). This stands in contrast with findings in the literature that better ratings are correlated with shorter durations. Although the model is not sufficiently strong to lend much confidence in this finding, the model interactions suggest that longer planned activities are more cost-effective.

Cost-effectiveness outcomes were found to require more work to extract sufficient data for reliable outcome prediction. Furthermore, obtaining cost-effectiveness metrics was found to be more challenging than obtaining ratings, in contrast to the claims from prior literature (Goldemberg, Jordan, and Kenyon, 2025).

It was also found that failing to divide by the total disbursement as marked in IATI increases predictability. This is in part because there is some noise inherent in the total expenditure, and also because when not dividing by expenditure, the model learns to predict linearly higher cost-effectiveness outcomes correlating with the expenditure.

Overall, little can currently be decisively concluded from the cost-effectiveness outcome analysis. For more detailed work, expert coding is likely required for robust extraction of quantitative outcomes, and funding breakdowns for projects should be used to more accurately evaluate cost-effectiveness, rather than the coarse assumption that all funding for a project goes to all outcomes. New water or sanitation piping connections is a more clearly comparable outcome, although there may be systematic differences in the costs of

Table 3: Predictive performance of cost-effectiveness outcome RF model on the validation set. R^2, Pairwise ordering probability, and Spearman correlation are shown with 95% bootstrap confidence intervals. In general, only the aggregate had sufficient data to produce a meaningfully predictive model, although the model is only able to distinguish more cost-effective activities 60% of the time. Cost-effectiveness outcomes are sorted by R^2 (descending). The Benefit-Cost Ratios had identical forecasts for all examples, hence a pairwise and Spearman could not be properly calculated. All other cost-effectiveness outcomes had fewer than 10 items in training or validation. 95% confidence intervals are determined via bootstrap over the prediction set, which was only computed once.

Outcome	R^2	Pairwise (%)	Spearman	MAE	$N_{\text{train}}/N_{\text{val}}/N_{\text{test}}$
All selected outcomes (z-aggregate activity mean)	0.05 [-0.03, 0.13]	60 [54, 67]	0.31 [0.13, 0.46]	0.56	682 / 299 / 299
Water Connections Connections ($log_{10}(connections)$)	-0.01 [-0.39, 0.13]	66 [54, 77]	0.43 [0.07, 0.69]	0.70	71 / 25 / 25
Economic Rate Of Return Percent (percent)	-0.01 [-0.19, 0.03]	53 [45, 60]	0.09 [-0.14, 0.28]	15.39	293 / 103 / 103
CO_2 Emission Reductions Tonnes CO2E ($log_{10}(tonnes\ CO_2e)$)	-0.03 [-0.32, 0.00]	41 [27, 58]	-0.28 [-0.62, 0.18]	1.46	38 / 23 / 23
Benefit Cost Ratios Ratio (ratio)	-0.05 [-0.42, -0.00]	N/A	N/A	0.33	44 / 18 / 18
Financial Rate Of Return Percent (percent)	-0.15 [-1.19, 0.03]	51 [39, 64]	0.01 [-0.32, 0.36]	15.20	147 / 43 / 43
Generation Capacity ($log_{10}(Mw)$)	-0.32 [-1.13, 0.02]	62 [47, 75]	0.39 [-0.06, 0.68]	0.73	44 / 20 / 20

sanitation connections and water supply that are not disambiguated by the model.

Despite these limitations, it appears that even a coarse coding of directly comparable activity cost-effectiveness outcomes is likely to provide a more robust ordering of overall activity cost-effectiveness than evaluator ratings alone, although cost-effectiveness outcome predictions are not statistically significantly better than forecasting the mean value for the cost-effectiveness outcome in a given sector.

4 Conclusion

The findings regarding each research question posed in the introduction are summarized below.

How do judgemental forecasting methods compare to statistical models in forecasting international aid overall success ratings and quantitative outcomes? LLM forecasts themselves (even when including a full RAG- and KNN-augmented multi-

stage prompting pipeline) consistently underperform statistical models in this domain, while having the disadvantage of being costlier and more difficult to iterate with. However, they provide meaningfully accurate narrative forecasts when combined with statistical models.

How do differing methods of combining judgemental and statistical forecasting compare in this domain? Embeddings of language models inserted into statistical models significantly improve forecasting ability. Furthermore, using embeddings as a means of selecting nearest neighbors and as a component of RAG retrieval gathered information that reliably increased the similarity grades between LLM forecasts and the eventual outcome over all tested LLM forecast configurations. Embeddings were also effective as a method for clustering activity disbursements as measured by improved performance on the validation set.

Language models themselves were helpful in extracting grades for various aspects of the forecast, which as a group improved the statistical model forecast. While direct averaging did not demonstrate improved performance, it was found that training a simple model to use the residual between the LLM and statistical model modestly improved forecasting skill across most metrics. There is some evidence that training a secondary model to incorporate the final LLM prediction with a RF prediction can also improve the overall forecast skill.

What methods improve the accuracy of narrative (qualitative) forecasts in this domain? It appears that the only reliable way of improving ratings over a range of different configurations was 1. to switch the model from the somewhat smaller *deepseek-V3.2* or *gemini-2.5-flash* models to the more expensive but more generally capable *gemini-3-pro* model, 2. to incorporate additional retrieval of information via RAG and inserting outcomes from similar past activities using the KNN technique and 3. to directly prompt the model to come to the same conclusion as the RF model.

Additional prompts to elicit reasons the outcome may go well or badly (stages 1 and 2 in the methods) sometimes deteriorated scores in certain model configurations, although the best performing models used these stages in its context window when forecasting. Similarly, fine-tuning sometimes helped, and sometimes deteriorated scores.

What aspects of the activity available in the dataset at the beginning of the activity lead to higher or lower ratings? As others in the literature have reported, decreased duration and increased planned expenditures tend to correlate with higher activity ratings (Vivalt, 2020) (L. Ashton et al., 2023) (Eilers et al., 2025). However, there is moderate evidence from the cost-effectiveness RF model that cost-effectiveness of quantitative outcomes does not clearly correlate with increased spending, and that increased durations also increase cost-effectiveness. Rating organizations tend to differ systematically, and by-sector differences in ratings appear to be more significant in general

than regional differences. However, this was not the core focus of this work and needs further research.

How does forecasting aggregate cost-effectiveness activity outcomes compare to forecasting ratings? The current dataset appears insufficient to forecast activity outcomes, both in numbers of quantitative outcomes and noisiness inherent in their extraction. There is some evidence in the validation set that an aggregate measure of cost-effectiveness can be used to rank promising activities, with a 60% chance of a correct ordering compared to 50% which would be arrived at by random chance.

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Declaration of AI Use

ChatGPT 5 and Claude Sonnet were used occasionally to write individual sentences in the Methods section summarizing code and to format equations in LaTeX. All LLM-authored text was carefully checked for validity. ChatGPT 5.2 and Claude Sonnet were also used to provide feedback for spelling, writing style and grammar.

Potsdam, 24 March 2026